

OGSTools: A Python package for OpenGeoSys

Tobias Meisel^{1,4}, Florian Zill^{2,1*}, Julian Heinze^{1*}, Lars Bilke^{1*}, Max Jäschke^{3,1,4*}, Feliks Kiszurno^{2,1*}, Norbert Grunwald^{1*}, Thomas Fischer^{1*}, and Jörg Buchwald^{1*}

¹ Helmholtz Centre for Environmental Research, Germany ² TU Bergakademie Freiberg, Germany ³ Leipzig University of Applied Sciences, Germany ⁴ Technische Universität Dresden, Germany * These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Editor: [Andrew Walker](#)

Reviewers:

- [@daavid00](#)
- [@jlarrahondo](#)

Submitted: 13 February 2026

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

OGSTools (OpenGeoSys Tools) is a Python library for pre- and post-processing of OpenGeoSys 6 (OGS) — a software package for simulating thermo-hydro-mechanical-chemical (THMC) processes in porous and fractured media (Bilke, Naumov, et al., 2025; Kolditz et al., 2012). OGSTools (Meisel et al., 2025) provides an interface between OGS-specific data and well-established data structures of the Python ecosystem, as well as domain-specific solutions, and examples for OGS users and developers. The library's functionalities are designed to be used in the OGS benchmark gallery, the OGS test suite, and for automating repetitive tasks in the model development cycle — from simple daily tasks to complex automated workflows. Figure 1 summarises these capabilities graphically.

Statement of need

Development efficiency

Modellers of OGS iteratively run simulations, analyse results, and refine their models. To improve efficiency, repetitive steps in the model development cycle are formalised in Python, matching the existing expertise of the user base. The Python library introduced here serves as a central platform to collect and improve common functionalities needed by modellers of OGS.

Complex workflows

In our scientific research, workflows integrate multiple steps — geological data preprocessing, ensemble simulations with OGS, domain-specific analysis and visualisation — into fully automated, reproducible sequences. Workflow-based approaches adhere to the FAIR principles (Goble et al., 2020; Wilkinson et al., 2025), using workflow management software for dependency management, execution control, and data provenance (Bilke, Fischer, et al., 2025). Building on Python-based workflow managers like Snakemake (Köster & Rahmann, 2012) and AiIDA (Huber et al., 2020), OGSTools provides reusable, domain-specific functionality as the shared foundation across workflow components.

Test suite

OGSTools provides functionality for (1) setting up test environments, (2) executing OGS under specified conditions, (3) evaluating OGS results against defined test criteria, and (4) monitoring the testing process.

37 Educational Jupyter notebooks

38 OGS is well suited for academic courses and teaching environments. With Jupyter Notebooks,
39 students can explore interactive learning environments where they directly modify parameters,
40 material laws, and other influencing factors, and instantly visualise the outcomes. OGSTools
41 reduces the boilerplate and keeps notebooks focused on the learning objective.

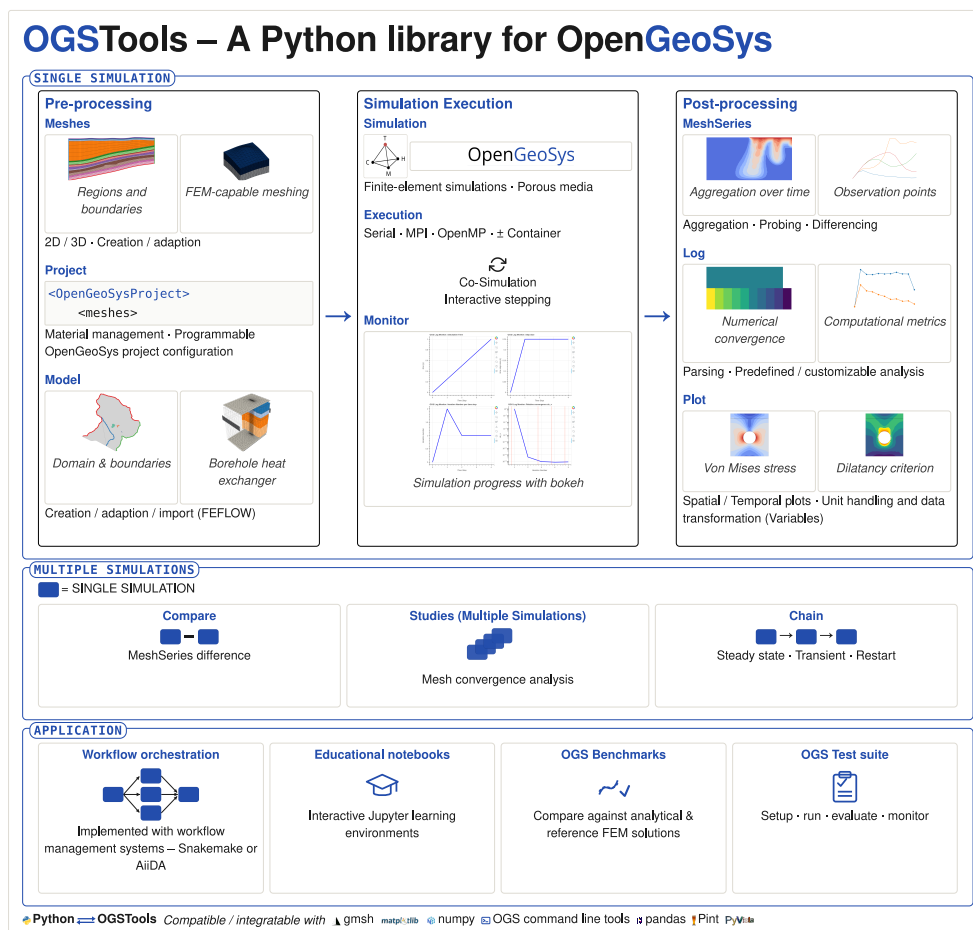


Figure 1: OGSTools graphical abstract: pre-processing, simulation execution, and post-processing for a single simulation; combining multiple simulations; and application areas.

42 State of the field

43 Simulator-specific companion libraries have emerged as a recurring pattern across scientific
44 computing domains. These software packages connect a domain-specific simulator with
45 the data structures and tooling of a general-purpose programming ecosystem (e.g. Python),
46 typically covering pre-processing, execution, and post-processing on a single programmatic
47 platform. OGSTools follows this pattern, in this case with OGS as the simulator.

48 In computational geoscience, several simulators have companion libraries of this kind. FloPy
49 (Hughes et al., 2024) wraps the MODFLOW family of groundwater flow and transport models,
50 supporting model creation, execution, and result analysis. pyGSFLOW (Larsen et al., 2022)
51 provides equivalent functionality for the GSFLOW integrated hydrologic model, and toughio
52 (Luu, 2020) covers pre- and post-processing for the TOUGH simulator family. Outside this
53 group, DOLFINx (Baratta et al., 2023) takes a different approach: it exposes FEM assembly

54 and solving directly through a Python API rather than wrapping an external solver.

55 An alternative to these scripting-based libraries is a dedicated companion GUI — as with
56 ModelMuse ([Winston, 2019](#)) for MODFLOW and the Data Explorer ([Rink et al., 2012](#)) for
57 OpenGeoSys.

58 Build vs. contribute

59 OGSTools contains only functionality that is explicitly specific to [OpenGeoSys](#) — domain-
60 specific data structures, OGS input/output formats, and process-specific defaults. General-
61 purpose functionality is deliberately left to established libraries (PyVista, Pandas, NumPy,
62 Pint), which OGSTools relies on.

63 Previously, without any centralisation to contribute OGS-specific pre- and postprocessing
64 code, the code base for Python-related tasks in OGS was fragmented, with components often
65 developed for specific use cases and varying degrees of standardisation, quality and maintenance
66 efforts. Further, OGSTools enables the transfer of years of experience in maintaining the OGS
67 core ([Bilke et al., 2019](#)) to the pre- and post-processing code. For the centralised approach,
68 preceding work on msh2vtu ([Kern & Bilke, 2022](#)), ogs6py and VTUInterface ([Buchwald et
69 al., 2021](#)) and further not yet published functionalities have been adapted and integrated into
70 OGSTools.

71 Software Design

72 Design choices

73 The functionality is grouped thematically into sub-libraries. Beyond general software engineering
74 best practices, the following design principles deserve particular attention.

75 **Open interfaces to common Python libraries:** Each sub-library either transforms OpenGeoSys
76 specific data into common Python data structures (e.g. PyVista, Pandas, NumPy, Matplotlib,
77 Pint), or vice versa. Users can use any subset of the library without lock-in, including when
78 preferring to run OpenGeoSys from the command line.

79 **Reuse OGS command line tools:** The new functionality combines the [OGS command line
80 tools](#) to cover more complex tasks than any single tool can handle alone.

81 **Fail loudly:** Silently producing wrong results is the highest risk in our simulation workflows.
82 OGSTools therefore raises errors immediately when constraints or plausibility checks are violated,
83 prioritising early failure over silent pass-through.

84 **Large data awareness:** The top-level API is designed to be approachable for small datasets at
85 entry level, while experts working with large data can tune performance via lower-level API
86 access or fall back to command-line tools and custom code.

87 **Infrastructure:** Code development is centralised on a self-hosted GitLab instance. A multi-
88 platform CI pipeline (Linux, Windows, macOS) enforces code quality and reports test coverage.
89 Examples are organised as plain Python scripts automatically converted to Jupyter notebooks,
90 with Binder integration for every release. To address the need for a versioned analysis
91 environment ([Blomer et al., 2014](#)), OGSTools ships a pinned dependency environment updated
92 with every release, while continuously tested against the latest dependencies.

93 Example

94 The following example shows a complete [OGS Liquid Flow](#) simulation workflow, adapted to 2D
95 from [an OGS benchmark](#). First, an OGS-capable mesh is generated and pressure boundary
96 conditions are assigned to the boundary meshes ([Figure 2](#)), using standard PyVista ([Sullivan
97 & Kaszynski, 2019](#)) functionality. After execution of the simulation, convergence metrics

98 (Figure 3) and the final pressure distribution (Figure 4) are visualised. An annotated version of
99 this example is available in the [OGSTools documentation](#).

```
import numpy as np
import ogstools as ot
from ogstools.examples import load_project_simple_lf

# 1. Pre-processing: Load example Project and construct input meshes
project = load_project_simple_lf()
meshes = ot.Meshes.from_gmsh(ot.gmsh_tools.rect((8,4),8,2))
# Set boundary conditions on the pyvista meshes
meshes["left"].point_data["pressure"] = 2.9e7
meshes["right"].point_data["pressure"] = 3.1e7
model = ot.Model(project=project, meshes=meshes)
# Visualize setup with boundary conditions (Figure 2)
model.plot_constraints()

# 2. Run: Execute Simulation
sim = model.run()

# 3. Post-processing: Analyse results

# Plot convergence behaviour (Figure 3)
sim.log.plot_convergence()

# Plot final pressure distribution (Figure 4)
ot.plot.contourf(sim.meshseries[-1], "pressure")

# 4. Store: Save Simulation
sim.save(id = "mysim", archive=True)
```

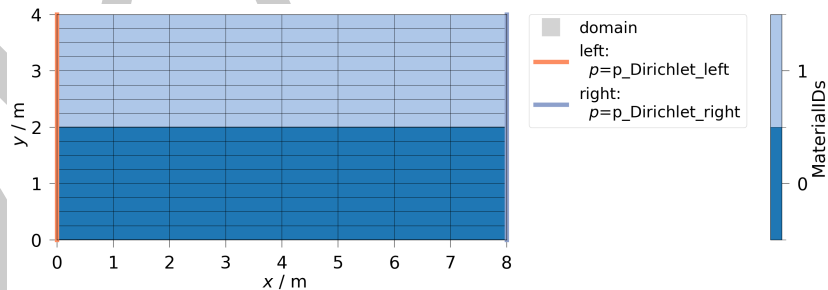


Figure 2: Initial boundary conditions.

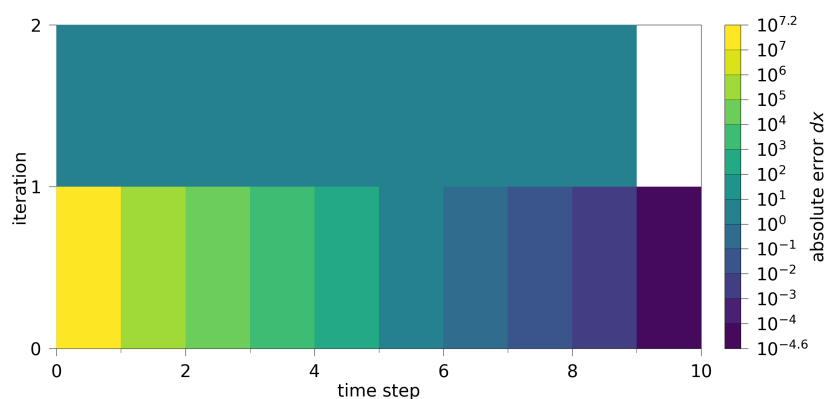


Figure 3: Resulting convergence metrics.

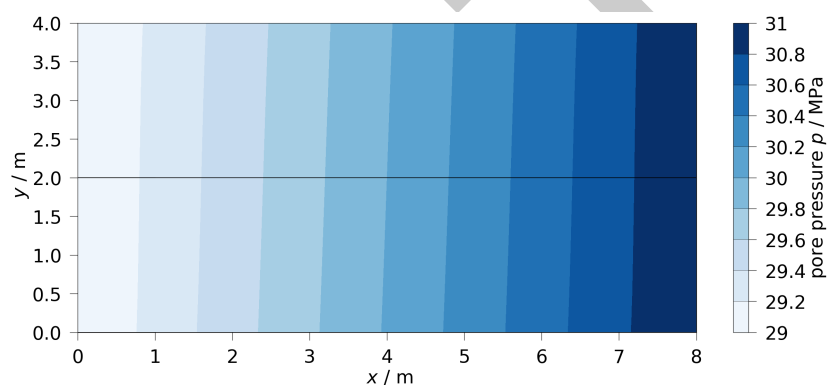


Figure 4: Resulting pressure distribution.

The example is deliberately kept minimal to keep the code listing short. OGSTools handles considerably more complex workflows, e.g. with more elaborate geometries or coupled physical processes, as shown by the OGS benchmarks for the [GREAT cell experiment suite](#), [excavation under two-phase flow](#), and [Kirsch's problem](#).

Features

The implemented features cover pre-processing, setup and execution of simulations, and post-processing.

Pre-processing for OGS includes mesh creation, adaptation, conversion, as well as defining boundary conditions, source terms, and generating project files (OGS-specific XML files). OGSTools further provides a material management component that allows process-specific material definitions to be assembled from structured YAML sources and translated into OGS-compatible project file entries. This allows a consistent, database-like handling of material parameters across workflows, test cases, and educational examples, while separating physical model definitions from project file syntax. In addition, a FEFLOW converter (from FEFLOW models to OGS models) is integrated ([Heinze et al., 2025](#)).

The simulation execution part covers running simulations with the OGS core via the command line and Python-based co-simulation interfaces. Runtime features include monitoring, interactive stepping, and access to intermediate results for in-simulation analysis.

Post-processing includes domain-specific evaluation and visualisation of simulation results for temporal and spatial distribution analysis, with sensible defaults and OGS-specific standards for plotting, and comparison against experimental data or analytical solutions.

The complete feature list is found in the [online documentation](#). Containers are provided for reproducibility, benefiting both developers and users (Bilke, Fischer, et al., 2025). Like OpenGeoSys, OGSTools is available via PyPI and conda-forge.

Research impact

Workflows

OGSTools emerged from and is used in several research projects. The AREHS-Project (Kahnt et al., 2021) is focused on modelling the effects of the glacial cycle on hydro-geological parameters in potential geological nuclear waste repositories in Germany. Within this project, Zill et al. (2024) and Silbermann et al. (2025) conducted their work using automated OGSTools workflows, with all material available at Meisel et al. (2024). OpenWorkFlow (Lehmann et al., 2024) is a project for an open-source, modular synthesis platform designed for safety assessment in the nuclear waste site selection procedure of Germany. ThEDi, a completed study on optimal disposal container packing to meet repository temperature limits, is one of multiple studies within OpenWorkFlow, mostly implemented using OGSTools.

OpenGeoSys benchmarks

The OGS benchmark gallery is a collection of web documents (mostly generated from Jupyter Notebooks) that demonstrate how users can set up, adjust, execute, and analyse simulations. They are well-suited as a starting point of research, and can be downloaded, executed, and adapted interactively. With OGSTools, code complexity and code duplication have been reduced, allowing especially inexperienced users to focus on the important part of the notebook.

Acknowledgements

This work has been supported by multiple funding sources, including SUTOGS (Streamlining Usability and Testing of OpenGeoSys) under [grant 528785032](#) by Deutsche Forschungsgemeinschaft (DFG), OpenWorkFlow under grant STAFuE-21-05-Klei by Bundesgesellschaft für Endlagerung (BGE), AREHS under grant 4719F10402 by Bundesamt für die Sicherheit der nuklearen Entsorgung (BASE), and DigBen under grant 03G0927A by Bundesministerium für Forschung, Technologie und Raumfahrt (BMFTR).

AI usage disclosure

In preparing this manuscript, the authors used Anthropic's Claude and OpenAI's ChatGPT, predominantly for grammar and spelling corrections, which the authors reviewed. For contributions to the software itself, the use of LLMs is permitted. Contributors are currently not required to disclose it; all contributions instead pass through a review process by the developers, maintainers, and authors.

References

- Baratta, I. A., Dean, J. P., Dokken, J. S., Habera, M., Hale, J. S., Richardson, C. N., Rognes, M. E., Scroggs, M. W., Sime, N., & Wells, G. N. (2023). *DOLFINx: The next generation FEniCS problem solving environment*. Zenodo. <https://doi.org/10.5281/zenodo.10447666>
- Bilke, L., Fischer, T., Naumov, D., & Meisel, T. (2025). Reproducible HPC software deployments, simulations, and workflows – a case study for far-field deep geological repository assessment. *Environmental Earth Sciences*, 84(17), 502. <https://doi.org/10.1007/s12665-025-12501-z>
- Bilke, L., Flemisch, B., Kalbacher, T., Kolditz, O., Helmig, R., & Nagel, T. (2019). Development of Open-Source Porous Media Simulators: Principles and Experiences. *Transport in Porous Media*, 130(1), 337–361. <https://doi.org/10.1007/s11242-019-01310-1>

- 165 Bilke, L., Naumov, D., Wang, W., Fischer, T., Kiszkurno, F. K., Lehmann, C., Max, J.,
166 Zill, F., Buchwald, J., Grunwald, N., Kessler, K., Aubry, L., Dörnbrack, M., Nagel, T.,
167 Ahrendt, L., Kaiser, S., & Meisel, T. (2025). *OpenGeoSys* (Version 6.5.4). Zenodo.
168 <https://doi.org/10.5281/zenodo.14672997>
- 169 Blomer, J., Berzano, D., Buncic, P., Charalampidis, I., Ganis, G., Lestaris, G., & Meusel, R.
170 (2014). The need for a versioned data analysis software environment. *CoRR*, *abs/1407.3063*.
171 <https://doi.org/10.48550/arXiv.1407.3063>
- 172 Buchwald, J., Kolditz, O., & Nagel, T. (2021). ogs6py and VTUinterface: Streamlining
173 OpenGeoSys workflows in python. *Journal of Open Source Software*, *6*(67), 3673. <https://doi.org/10.21105/joss.03673>
174
- 175 Goble, C., Cohen-Boulakia, S., Soiland-Reyes, S., Garijo, D., Gil, Y., Crusoe, M. R., Peters, K.,
176 & Schober, D. (2020). FAIR computational workflows. *Data Intelligence*, *2*(1-2), 108–131.
177 https://doi.org/10.1162/dint_a_00033
- 178 Heinze, J., Lehmann, C., Meisel, T., Rink, K., Kreye, P., Renz, A., Zeunert, S., & Rühaak, W.
179 (2025). Combining FEFLOW and OpenGeoSys for interoperable workflows in environmental
180 geotechnics. *Environmental Earth Sciences*, *84*(16), 457. <https://doi.org/10.1007/s12665-025-12380-4>
181
- 182 Huber, S. P., Zoupanos, S., Uhrin, M., Talirz, L., Kahle, L., Häuselmann, R., Gresch, D., Müller,
183 T., Yakutovich, A. V., Andersen, C. W., Ramirez, F. F., Adorf, C. S., Gargiulo, F., Kumbhar,
184 S., Passaro, E., Johnston, C., Merkys, A., Cepellotti, A., Mounet, N., ... Pizzi, G. (2020).
185 AiiDA 1.0, a scalable computational infrastructure for automated reproducible workflows
186 and data provenance. *Scientific Data*, *7*(1). <https://doi.org/10.1038/s41597-020-00638-4>
- 187 Hughes, J. D., Langevin, C. D., Paulinski, S. R., Larsen, J. D., & Brakenhoff, D. (2024). FloPy
188 workflows for creating structured and unstructured MODFLOW models. *Groundwater*,
189 *62*(1), 124–139. <https://doi.org/10.1111/gwat.13327>
- 190 Kahnt, R., Konietzky, H., Nagel, T., Kolditz, O., Jockel, A., Silbermann, C., Tiedke,
191 F., Meisel, T., Rink, K., Wang, W., Zill, F., Carl, A., Gabriel, A., Schlegel, M.,
192 & Abraham, T. (2021). AREHS: Effects of changing boundary conditions on the
193 development of hydrogeological systems: Numerical long-term modelling considering
194 thermal–hydraulic–mechanical(–chemical) coupled effects. *Safety of Nuclear Waste*
195 *Disposal*, *1*, 175–177. <https://doi.org/10.5194/sand-1-175-2021>
- 196 Kern, D., & Bilke, L. (2022). msh2vtu. In *GitHub repository*. GitHub. <https://github.com/dominik-kern/msh2vtu>
197
- 198 Kolditz, O., Bauer, S., Bilke, L., Grunwald, N., Delfs, J.-O., Fischer, T., Görke, U., Kalbacher,
199 T., Kosakowski, G., McDermott, C., Park, C.-H., Radu, F., Rink, K., Shao, H., Sun,
200 F., Sun, Y., Singh, A., Taron, J., Walther, M., & Zehner, B. (2012). OpenGeoSys:
201 An open-source initiative for numerical simulation of thermo-hydro-mechanical/chemical
202 (THM/c) processes in porous media. *Environmental Earth Sciences*, *67*, 589–599. <https://doi.org/10.1007/s12665-012-1546-x>
203
- 204 Köster, J., & Rahmann, S. (2012). Snakemake—a scalable bioinformatics workflow engine.
205 *Bioinformatics*, *28*(19), 2520–2522. <https://doi.org/10.1093/bioinformatics/bts480>
- 206 Larsen, J. D., Alzraiee, A., & Niswonger, R. G. (2022). Integrated hydrologic model
207 development and postprocessing for GSFLOW using pyGSFLOW. *Journal of Open Source*
208 *Software*, *7*(72), 3852. <https://doi.org/10.21105/joss.03852>
- 209 Lehmann, C., Bilke, L., Buchwald, J., Graebing, N., Grunwald, N., Heinze, J., Meisel, T., Lu,
210 R., Naumov, D., Rink, K., Sen, O. Ö., Selzer, P., Shao, H., Wang, W., Zill, F., Nagel,
211 T., & Kolditz, O. (2024). OpenWorkFlow—development of an open-source synthesis-
212 platform for safety investigations in the site selection process. *Grundwasser*, *29*(1), 31–47.
213 <https://doi.org/10.1007/s00767-024-00566-9>

- 214 Luu, K. (2020). Toughio: Pre- and post-processing python library for TOUGH. *Journal of*
215 *Open Source Software*, 5(51), 2412. <https://doi.org/10.21105/joss.02412>
- 216 Meisel, T., Zill, F., Jäschke, M., Bilke, L., Grunwald, N., Fischer, T., & Kiszkurno, F. K.
217 (2025). *OGSTools* (Version 0.7.1). Zenodo. <https://doi.org/10.5281/zenodo.17223939>
- 218 Meisel, T., Zill, F., Silbermann, C. B., Wang, W., Bilke, L., & Kern, D. (2024). *AREHS -*
219 *OpenGeoSys workflow* (Version 1.0). Zenodo. <https://doi.org/10.5281/zenodo.11367280>
- 220 Rink, K., Kalbacher, T., & Kolditz, O. (2012). Visual data exploration for hydrological analysis.
221 *Environmental Earth Sciences*, 65(5), 1395–1403. [https://doi.org/10.1007/s12665-011-](https://doi.org/10.1007/s12665-011-1230-6)
222 [1230-6](https://doi.org/10.1007/s12665-011-1230-6)
- 223 Silbermann, C., Zill, F., Meisel, T., Kern, D., Kolditz, O., Magri, F., & Nagel, T. (2025).
224 Automated thermo-hydro-mechanical simulations capturing glacial cycle effects on nuclear
225 waste repositories in clay rock. *Geomechanics and Geophysics for Geo-Energy and Geo-*
226 *Resources*, 11. <https://doi.org/10.1007/s40948-025-00960-4>
- 227 Sullivan, C., & Kaszynski, A. (2019). PyVista: 3D plotting and mesh analysis through a
228 streamlined interface for the visualization toolkit (VTK). *Journal of Open Source Software*,
229 4(37), 1450. <https://doi.org/10.21105/joss.01450>
- 230 Wilkinson, S. R., Aloqalaa, M., Belhajjame, K., Crusoe, M. R., Paula Kinoshita, B. de, Gadelha,
231 L., Garijo, D., Gustafsson, O. J. R., Juty, N., Kanwal, S., Khan, F. Z., Köster, J., Peters-von
232 Gehlen, K., Pouchard, L., Rannow, R. K., Soiland-Reyes, S., Soranzo, N., Sufi, S., Sun, Z.,
233 ... Goble, C. (2025). Applying the FAIR principles to computational workflows. *Scientific*
234 *Data*, 12(1). <https://doi.org/10.1038/s41597-025-04451-9>
- 235 Winston, R. B. (2019). *ModelMuse version 4 — a graphical user interface for MODFLOW*
236 *6* (Scientific Investigations Report No. 2019-5036). U.S. Geological Survey. <https://doi.org/10.3133/sir20195036>
- 237
- 238 Zill, F., Silbermann, C., Meisel, T., Magri, F., & Nagel, T. (2024). Far-field modelling of
239 THM processes in rock salt formations. *Open Geomechanics*, 5, 1–16. [https://doi.org/10.](https://doi.org/10.5802/ogeo.20)
240 [5802/ogeo.20](https://doi.org/10.5802/ogeo.20)